AP BIOLOGY TOPIC 4.2

Introduction to Signal Transduction: Reception, Transduction & Response

UNIT 4: CELL COMMUNICATION & CELL CYCLE (10–15% EXAM WEIGHT)

IST-3.C | IST-3.D | GPCRS, RTKS & SECOND MESSENGERS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 4.2 (INTRODUCTION TO SIGNAL TRANSDUCTION: RECEPTION, TRANSDUCTION & RESPONSE)

CED Core Vocabulary

Signal Transduction Pathway — The molecular mechanism that converts an extracellular chemical signal (ligand) into a specific intracellular response across 3 sequential stages : Reception:...

Response — Transduced signal triggers a specific cellular activity (e.g., gene expression/transcription, enzyme activation, ion channel opening, secretion, or cell division).

Mechanism — Extracellular ligand binds GPCR → GPCR changes shape and activates an intracellular G protein (swaps inactive GDP for active GTP) → active G protein slides...

Phosphorylation Cascades ("Domino Effect") — A series of relay proteins activated sequentially by adding phosphate groups from ATP.

Transduction — The receptor's shape change initiates a multistep intracellular cascade (via phosphorylation cascades and second messengers) that relays and amplifies the signal.

1. G Protein-Coupled Receptors (GPCRs) — Largest family of cell-surface receptors in eukaryotes (7 transmembrane α -helices).

3. Ligand-Gated Ion Channels — Transmembrane pore proteins that open or close a gate when a specific ligand binds.

Protein Kinases — Enzymes that transfer PO_4^{3-} from ATP to the next relay protein → changes its 3D shape (usually turning it ON).

§2. THE 3 STAGES OF SIGNAL TRANSDUCTION & LIGAND CHEMISTRY

Reception → Transduction → Response

- **Signal Transduction Pathway:** The molecular mechanism that converts an extracellular chemical signal (**ligand**) into a specific intracellular response across **3 sequential stages**:

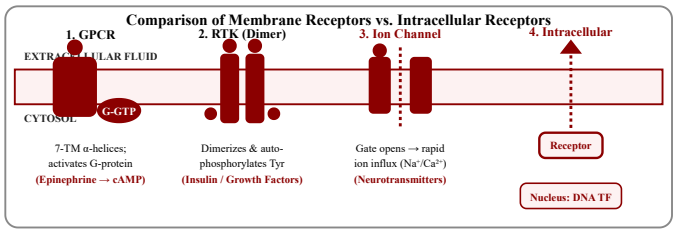
1. **Reception:** Target cell detects a signaling molecule (**ligand**) when it binds to a specific receptor protein, causing an immediate **conformational (3D shape) change** in the receptor.
2. **Transduction:** The receptor's shape change initiates a multistep intracellular cascade (via **phosphorylation cascades** and **second messengers**) that relays and **amplifies** the signal.
3. **Response:** Transduced signal triggers a specific cellular activity (e.g., **gene expression/transcription, enzyme activation, ion channel opening, secretion, or cell division**).

FEATURE	CELL-SURFACE (MEMBRANE) RECEPTORS	INTRACELLULAR (CYTOPLASMIC/NUCLEAR) RECEPTORS
Ligand Properties	**Hydrophilic (polar, water-soluble)** or large molecules — **CANNOT cross** the nonpolar phospholipid bilayer core.	**Hydrophobic (nonpolar, lipid-soluble)** or very small gases — **diffuse directly across** the plasma membrane.
Ligand Examples	Peptide/protein hormones (**Insulin**, growth factors), amine hormones (**Epinephrine**), neurotransmitters.	**Steroid hormones** (testosterone, estrogen, cortisol, aldosterone), **thyroid hormone**, nitric oxide (**NO**).
Mechanism	Ligand stays outside cell; binding extracellular domain changes intracellular domain shape → second messengers.	Hormone-receptor complex enters nucleus and acts directly as a **transcription factor** binding DNA to turn genes ON/OFF.

[COMMON AP EXAM FRQ TRAP] **Crucial Concept Check:** For cell-surface receptors, **the hydrophilic ligand NEVER enters the cell!** It binds to the extracellular face of the receptor, and the *information* is passed inward via a **conformational shape change** in the receptor protein.

§3. 4 MAJOR RECEPTOR MECHANISMS VISUALIZED

GPCR, RTK, Ion Channel & Intracellular



§4. 3 CLASSES OF CELL-SURFACE MEMBRANE RECEPTORS

Hydrophilic Ligands

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

3 Classes of Cell-Surface Membrane Receptors ➔

1. G Protein-Coupled Receptors (GPCRs) ➔ **3. Ligand-Gated Ion Channels**

- **1. G Protein-Coupled Receptors (GPCRs):**
  - Largest family of cell-surface receptors in eukaryotes (7 transmembrane **α**-helices).
  - **Mechanism:** Extracellular ligand binds GPCR → GPCR changes shape and activates an intracellular **G protein** (swaps inactive **GDP for active GTP**) → active G protein slides along membrane to activate an effector enzyme (like **adenylyl cyclase**) → hydrolyzes GTP back to GDP to turn itself off.
  - *Example:* **Epinephrine (adrenaline)** binding liver/muscle GPCRs to trigger glycogen breakdown.
- **2. Enzyme-Linked Receptors / Receptor Tyrosine Kinases (RTKs):**
  - Membrane receptors with an intracellular enzymatic kinase domain.
  - **Mechanism:** Ligand binding causes two RTK monomers to pair up (**dimerization**) → cytoplasmic tails **autophosphorylate** multiple tyrosine amino acids using ATP → simultaneously activates **multiple downstream relay pathways at once** (critical for cell growth/mitosis!).
  - *Examples:* **Insulin receptor**, growth factor receptors (PDGF, EGF, FGF).
- **3. Ligand-Gated Ion Channels:**
  - Transmembrane pore proteins that open or close a gate when a specific ligand binds.
  - **Mechanism:** Ligand binds → gate opens → rapid flow of specific ions (**Na^+**, **K^+**, **Ca^{2+}**, **Cl^-**) down their electrochemical gradient → immediate change in membrane potential.
  - *Example:* **Acetylcholine (ACh)** opening **Na^+** channels at a synapse.

§5. TRANSDUCTION: CASCADES, SECOND MESSENGERS & AMPLIFICATION

Domino Effect

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Transduction ➔ **Phosphorylation Cascades ("Domino Effect")** ➔ **Second Messengers** ➔ **Signal Amplification** ➔ **Cell-Specific Responses**

- **Phosphorylation Cascades ("Domino Effect"):**
  - A series of relay proteins activated sequentially by adding phosphate groups from ATP.
  - **Protein Kinases:** Enzymes that transfer **PO_4^{3-}** from ATP to the next relay protein → changes its 3D shape (usually turning it **ON**).
  - **Protein Phosphatases:** Enzymes that remove **PO_4^{3-}** (dephosphorylation) → **inactivates** kinases and turns the signal pathway **OFF** when the ligand leaves.
- **Second Messengers:** Small, non-protein, water-soluble molecules or ions that rapidly diffuse through the cytosol to relay and amplify signals received by GPCRs or RTKs:
  - **Cyclic AMP (cAMP):** Synthesized from ATP by **adenylyl cyclase**; activates **Protein Kinase A (PKA)**. Converted back to inactive AMP by *phosphodiesterase (PDE)*.
  - **Calcium Ions (Ca^{2+}) & IP_3 / DAG: IP_3** opens ligand-gated **Ca^{2+}** channels on the Smooth ER, flooding the cytosol with **Ca^{2+}** to trigger muscle contraction or vesicle exocytosis.
- **Signal Amplification:** Because each enzyme in a cascade activates hundreds of substrate molecules before being turned off, **1 single hormone molecule binding 1 receptor can generate millions of active product molecules** inside the cell!
- **Cell-Specific Responses:** The **same ligand can produce completely different responses in different target cells** if those cells express different receptors, different intracellular relay proteins, or different transcription factors (e.g., epinephrine causes liver cells to break down glycogen, heart muscle to beat faster, and digestive blood vessels to constrict).

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 4.2 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 4.2: Introduction to Signal Transduction: Reception, Transduction & Response ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 (Mitosis in Onion Root Tips & Lectin Signaling): Root tip squash stained with aceto-orcein/toluidine blue to compare mitotic index (% dividing cells = [cells in PMAT / total cells counted] × 100) between control meristem and phytohemagglutinin (lectin) or caffeine-treated roots. Chi-square (χ^2) tests whether observed mitotic vs. interphase counts differ significantly from control. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Mitotic Index = (N_mitosis / N_total) × 100. DNA Content (C) & Ploidy (n) tracking in diploid (2n) somatic cells: G_1 = 2n, 2C (unreplicated chromosomes); S phase = DNA synthesis (2n, 2C → 4C); G_2/Prophase/Metaphase = 2n, 4C (sister chromatids joined); Anaphase = transient 4n, 4C (chromatids separate into individual chromosomes); Post-Cytokinesis = 2n, 2C. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Signaling Disruption Case Study: Vibrio cholerae secretes Cholera Toxin, which enzymatically modifies the G_α subunit of intestinal GPCRs so it cannot hydrolyze GTP → GDP. Locked in its active GTP-bound state, G_α continuously activates adenylyl cyclase, causing runaway cAMP accumulation, constitutive PKA activation, CFTR Cl^-/H_2O efflux, and life-threatening secretory diarrhea. In cell cycle control, loss-of-function mutations in both alleles of TP53 ('guardian of the genome') prevent p21-mediated G_1/S arrest and apoptosis of DNA-damaged cells, driving oncogenic proliferation.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 4.2 to prove why the claim holds.

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